

G&S 2.0 - Backend Specifications

G&S 2.0 — Scheduled Billable API: Backend Technical Reference

Created: July 13, 2026 | Last Modified: July 13, 2026

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1. Overview

Purpose: Send Goods & Services (G&S) charges from the Water Desk to Aspire as Scheduled Billables via a single POST call per submission.

Architecture: One Call URL action on the Dealer Goods & Services table fires per-submission, combining all child Fee records into a single API payload using formula columns and summary roll-ups.

Property	Value
Database	CSP (ApplID 76449) — test environment
Endpoint	POST /1/contract/scheduledbillables
Base URL	https://purewaterpartnerstest2.lease team.net/LeaseTeam.Aspire.Api
Auth	Bearer token (Aspire API key)

2. Data Flow

Taylor enters fees in Fees sub-tab

↓

```

Each fee's [API Asset Fragment] formula auto-computes JSON chunk
↓
G&S parent [API Assembled RelatedAssets] summary concatenates all chunks
G&S parent [Total Fee Amount] summary sums amounts
G&S parent [API First Fee *] summaries pull shared fields
G&S parent [API DoNotRelateAssets Value] formula computes true/false
↓
Taylor clicks "Send Scheduled Billable Test" button on G&S Preview Page
↓
Action 1: Send Scheduled Billable to Aspire (Call URL) – POST to Aspire
Action 2: Mark Related Fees Sent (Update Record) – stamps Status="Sent" (fires after API)
↓
Aspire returns 200 OK with Result: "Commit" (or 422 with error details)

```

3. Button Configuration

Button: "Send Scheduled Billable Test"

Property	Value
Location	Preview Page
Table	Dealer Goods & Services
Filter	All records
Roles	Default Role
Assignments	None

Actions (ordered):

Order	Action Name	Type	What It Does
1	Send Scheduled Billable to Aspire	Call URL	POST payload to Aspire endpoint
2	Mark Related Fees Sent (Parent Test)	Update Record	Sets related Fees records Status = "Sent"

Note: Action order was corrected on 07/13/2026. Call URL fires first; fees are only marked "Sent" after the API call completes.

4. API Endpoint

Method	URL	Headers	Body
POST	<code>https:// purewaterpartnerstest2.lease team.net/ LeaseTeam.Aspire.Api/1/contract/ scheduledbillables</code>	Authorization: Bearer {token}, Content-Type: application/json	See §5 below

5. Call URL Body (Exact — As Saved in TeamDesk)

Description

Constructs the Scheduled Billable JSON payload by pulling values from the G&S parent record's formula and summary columns.

Why This Step Exists

Aspire requires a single JSON object per Scheduled Billable submission. The payload combines header-level fields (ContractId, Description, TransactionCode) with per-asset detail (RelatedAssets array) rolled up from child Fee records.

What TeamDesk Does

On button click, TeamDesk evaluates the `<%= %>` and `<%=* %>` tokens against the current G&S record's column values, assembles the JSON, and sends the HTTP POST.

Call URL Body

```
{
  "RecordId": <%= [Ref#] %>,
  "Description": <%= [API First Fee Description] %>,
  "TransactionCode": {"Code": <%= [API First Fee TransCode] %>},
  "ContractId": {
    "Value": <%= [Contract RecordId] %>,
    "Type": "transaction"
  },
  "Payments": [
    {
      "StartDate": <%= Format([API First Fee Payment Date], "yyyy-MM-dd") %>,
      "Occurrences": <%= [API First Fee Occurrences] %>,
      "Frequency": <%= [API First Fee Frequency] %>,
      "Amount": <%= [Total Fee Amount] %>
    }
  ],
  "ProrationMethod": {
    "Code": 1,
    "Description": "EquipmentCost"
  },
  "RelatedAssets": [<%= [API Assembled RelatedAssets] %>],
  "DoNotRelateAssets": <%= [API DoNotRelateAssets Value] %>,
  "DelinquencyCode": null,
  "InvoiceCode": null,
  "InvoiceLeadDays": null,
  "SuspendInvoicing": true,
  "BillingType": null,
  "FollowRent": false
}
```

Result

Aspire creates a Scheduled Billable record and returns HTTP 200 with `"Result": "Commit"` on success, or HTTP 422 with error details on validation failure.

Token Type Reference

Token	Behavior
<%= %>	Auto-encodes value (wraps strings in quotes, leaves numbers raw)

Token	Behavior
<%* %>	Passthrough — inserts raw value with NO encoding (required for pre-computed JSON fragments and boolean literals)

6. Table: Fees (Child)

Purpose

Stores individual G&S charges. Each Fee record represents one line item that contributes to the parent G&S submission.

Input Columns

Column	Type	Notes
Equipment	Reference → Miscellaneous Units (DropDown)	Links to an asset; null = no asset
Description	Text (dropdown)	Ice Mats, Cleaning Fee, Trip Charge, etc.
Fee Amount	Numeric	Dollar amount for this line item
First Payment Date	Date	When billing begins
TransCode	Text	CLEANING, PM, LIFT, TRIP, FILTER, CLOG, CO2EXCH, ICEMAT
All Assets?	Checkbox	If true, Aspire prorates across all contract assets
Recurring Billable?	Checkbox	Enables multi-occurrence billing
Number of Occurrences	Numeric	How many billing cycles (used when Recurring = true)

Column	Type	Notes
Recurring Frequency	Text	Monthly, Quarterly, Annual

Lookup Columns (from Equipment → Miscellaneous Units)

Column	Source
Serial Number	Miscellaneous Units → Serial Number
Equipment ContractAssetId	Miscellaneous Units → ContractAssetId
Equipment Aspire Record Id	Miscellaneous Units → Aspire Record Id
Equipment Contract Asset Id Raw	Miscellaneous Units → Contract Asset Id Raw

Summary Columns (from Parent G&S)

Column	Summary Function	Source
API ContractId	Concatenate	Parent G&S → Selected ContractId
API Contract RecordId	Concatenate	Parent G&S → Contract RecordId
Parent Contract Status	Concatenate	Parent G&S → Contract Status
Parent Delinquency Code	Concatenate	Parent G&S → Delinquency Code
Parent Invoice Code	Concatenate	Parent G&S → Invoice Code

Key Formula: API Asset Fragment

Column Type: Formula - Text

Purpose: Builds the per-fee JSON object for the RelatedAssets array.

Why this exists: Each Fee with an equipment reference contributes one entry to the RelatedAssets array. This formula pre-computes the JSON fragment so the parent's summary column can concatenate them all.

```

If(
  IsNull([Equipment Aspire Record Id]),
  "",
  '{"AssetRecordId":{"Value":"' & JSONFormat([Equipment Aspire Record Id]) &
    '","Type":"record"},"PaymentAmount":"' & ToText([Fee Amount]) &
    '","FirstPaymentDate":"' & JSONFormat(XMLFormat([First Payment Date])) & '}'
)

```

Result: Produces a JSON object like:

```

{"AssetRecordId":
  {"Value":"APIE-93067","Type":"record"},"PaymentAmount":100,"FirstPaymentDate":"2026-06-29"}

```

Or empty string "" if no equipment is linked.

Other Formula Columns

Column	Type	Formula	Purpose
API Has Equipment	Checkbox	not IsNull([Equipment])	Guard check for asset-related logic
API DoNotRelateAssets	Checkbox	IsNull([Equipment]) and not [All Assets?]	True when charge has no asset association
API Frequency	Text	If([Recurring Billable?] and not IsNull([Recurring Frequency]), [Recurring Frequency], "Monthly")	Defaults to "Monthly" for one-time
API Occurrences	Numeric	If([Recurring Billable?] and [Number of Occurrences] > 0, [Number of Occurrences], 1)	Defaults to 1 for one-time
API Follow Rent Allowed	Checkbox	If([Follow Rent?] and [Parent Contract Status] =	Guard: prevents Follow Rent on

Column	Type	Formula	Purpose
		"Renewal", false, [Follow Rent?])	Renewal contracts

Response Columns

Column	Type	Purpose
Scheduled Billable API Response Status	Text (Updatable)	HTTP status code from Call URL response
Scheduled Billable API Response Body	Text (Updatable)	Full response body
Scheduled Billable API Error Message	Text (Updatable)	Parsed error detail
API Proration Method Description	Text (Updatable)	Confirmation of proration method used

7. Table: Dealer Goods & Services (Parent)

Purpose

Groups Fee records into a single billable submission. Holds the rolled-up summary data and fires the Call URL action.

Key Columns for API

Column	Type	Source/Formula	Purpose
Ref#	Autonumber	System-generated	Used as <code>RecordId</code> in payload (prefixed "REF#")
Contract RecordId	Lookup		Used as <code>ContractId.Value</code>

Column	Type	Source/Formula	Purpose
		Contract table → numeric Aspire contract ID	
Selected ContractId	Lookup	Contract table → contract number	Display only, NOT used in API

Summary Columns (Pulling from Child Fees)

Column	Summary Function	Sort	Source (Fees Table)	API Field
API Assembled RelatedAssets	Concatenate (separator: ,)	—	API Asset Fragment	<code>RelatedAssets</code> array contents
API First Fee Description	First	Id Asc	Description	<code>Description</code>
API First Fee TransCode	First	Id Asc	TransCode	<code>TransactionCode.Code</code>
API First Fee Occurrences	First	Id Asc	API Occurrences	<code>Payments[].Occurrences</code>
API First Fee Frequency	First	Id Asc	API Frequency	<code>Payments[].Frequency</code>
API First Fee Payment Date	First	Id Asc	First Payment Date	<code>Payments[].StartDate</code>
API First Fee AllAssets	First	Id Asc	All Assets?	Controls <code>DoNotRelateAssets</code> logic
API Units Count	# of records	—	Equipment (non-null only)	Controls <code>DoNotRelateAssets</code> logic
	Total	—	Fee Amount	<code>Payments[].Amount</code>

Column	Summary Function	Sort	Source (Fees Table)	API Field
Total Fee Amount				
API First Matching Fee Status	First	—	Status	Readiness check
API Related Fee Count	Count	—	Id	Readiness check

Key Formula: API DoNotRelateAssets Value

Column Type: Formula - Text

Purpose: Produces the literal string "true" or "false" for the DoNotRelateAssets field in the JSON payload.

Why this exists: TeamDesk's <%* %> passthrough token inserts this raw value directly into the JSON. A checkbox would output True / False (capitalized), which is invalid JSON.

```
If([API Units Count] = 0 and not [API First Fee AllAssets], "true", "false")
```

Logic:

- **true** when: No fees have equipment selected AND "All Assets?" is not checked (Scenario 1)
- **false** when: At least one fee has equipment, OR "All Assets?" is checked (Scenarios 2, 3, 5, 6)

Other Formula Columns

Column	Formula	Purpose
API ContractId Value	Nz([Selected ContractId])	Pulls ContractId for payload
	"transaction"	

Column	Formula	Purpose
API ContractId Type		Always "transaction"
API Ready for Single-Asset Test	not IsNull([Selected ContractId]) and [API Related Fee Count] > 0 and IsNull([API First Matching Fee Status])	Readiness guard
API Payload Preview	Status indicator for multi-asset grouping	Preview/debugging only
Aspire Http Link	Formula - Text	Constructs URL to view billable in Aspire

8. Aspire API Field Formats (Confirmed Working)

Field	Format	Example	Notes
TransactionCode	<code>{"Code": "<string>"}</code>	<code>{"Code": "CLEANING"}</code>	Valid codes: CLEANING, PM, LIFT, TRIP, FILTER, CLOG, CO2EXCH, ICEMAT
ContractId	<code>{"Value": "<numeric>", "Type": "transaction"}</code>	<code>{"Value": "104762", "Type": "transaction"}</code>	Value MUST be numeric string
AssetRecordId	<code>{"Value": "<string>", "Type": "record"}</code>	<code>{"Value": "APIE-93067", "Type": "record"}</code>	Uses Aspire Record Id format
ProrationMethod			Always present in current implementation

Field	Format	Example	Notes
	<code>{"Code": <int>, "Description": "<string>"}</code>	<code>{"Code": 1, "Description": "EquipmentCost"}</code>	
Dates	<code>yyyy-MM-dd</code>	<code>"2026-06-29"</code>	ISO format
Amounts	Raw numeric	<code>100</code>	No quotes, no currency symbols
FollowRent	Boolean literal	<code>false</code>	Always false (dropped from scope)
SuspendInvoicing	Boolean literal	<code>true</code>	Always true during rollout; flip to false after first week (per Cody, 06-30 meeting)
DoNotRelateAssets	Boolean literal	<code>true</code> or <code>false</code>	Computed by formula
RecordId	String	<code>"REF#49793"</code>	G&S Autonumber

9. Validation Rules (Learned from Testing)

These constraints were identified during test validation and will cause Aspire to return HTTP 422 if violated:

Rule	Error Behavior	Prevention
ContractId.Value MUST be numeric when Type="transaction"	422 Unprocessable Entity	Ensure <code>[Contract RecordId]</code> lookup resolves to a numeric value
		Validate dates in Fee entry form or via

Rule	Error Behavior	Prevention
Asset FirstPaymentDate MUST be >= Payments.StartDate	422 Unprocessable Entity	TeamDesk validation rule
TransactionCode.Code must exist in Aspire Admin billing configuration	422 Unprocessable Entity	Restrict dropdown values to known codes
Sum of RelatedAssets[].PaymentAmount must equal Payments[].Amount	422 Unprocessable Entity	[Total Fee Amount] summary ensures amounts sum correctly
TeamDesk <?%? %> conditional blocks do NOT reliably evaluate summary/formula columns	Malformed JSON sent to API	Use <?* %> passthrough with pre-computed formula columns instead

10. Validated Test Results (July 13, 2026)

Scenario	Description	RecordId	ContractId	DoNotRelateAssets	RelatedAssets	H St
1	No assets, one-time \$100	REF#49793	42712	true	[]	20
2	Single asset, \$100	REF#49794	42712	false	1 asset (APIE-93067)	20
3	Two assets, \$25/\$75 split	REF#49797	104762	false	2 assets (APIMU-161186, APIE-153343)	20
5	All assets proportional, \$100	REF#49798	104762	false	[] (Aspire prorates)	20

Scenario	Description	RecordId	ContractId	DoNotRelateAssets	RelatedAssets	H S
6	Recurring 3 months, 2 assets	REF#49799	104762	false	2 assets	20
4	Custom billing codes	—	—	—	—	Pe
7	Follow Rent, no assets	—	—	—	—	D
8	Follow Rent, 2 assets	—	—	—	—	D

Scenario Logic Matrix

Scenario	Equipment?	All Assets?	Recurring?	DoNotRelateAssets	RelatedAssets Content
1	No	No	No	true	Empty []
2	Yes (1)	No	No	false	1 asset with PaymentAmount + FirstPaymentDate
3	Yes (2+)	No	No	false	N assets, amounts sum to total
5	No	Yes	No	false	Empty [] — Aspire prorates across all
6	Yes (2+)	No	Yes	false	N assets, Occurrences > 1

11. Potential Issues

#	Issue	Impact	Mitigation
1	Multi-asset assumes same TransCode/Description	The "First" summaries pull TransCode and Description from the first fee only. If fees under one G&S parent have different TransCodes, only the first is sent to Aspire	Either enforce same TransCode per G&S submission (UI/validation), or split into separate API calls per TransCode group
2	Multi-asset assumes same First Payment Date	Payments.StartDate uses <code>API First Fee Payment Date</code> (first fee's date). If fees have different dates, asset FirstPaymentDates could fall outside the Payment date range, causing a 422 error	Either enforce same date per submission, or use the earliest date across all fees
3	No validation on Equipment Aspire Record Id	If a user selects equipment that hasn't been linked in Aspire (Aspire Record Id is empty), the asset fragment will be empty and the asset won't be included in the payload — silently dropped	Add a validation rule or form message: if Equipment is selected but <code>Equipment Aspire Record Id</code> is null, warn the user
4	FollowRent checkbox exists in schema but not on form	Column <code>Follow Rent?</code> exists on Fees table but is not displayed. If data gets set to true by accident (import, API, etc.), the per-fee payload formula could behave unexpectedly	Remove the FollowRent branch from formulas entirely, or delete/hide the column

12. Remaining Scenarios to Test

Scenario	Description	Status
4	Custom billing codes (DelinquencyCode, InvoiceCode, InvoiceLeadDays, BillingType)	Pending — requires fields for user to specify billing codes

Scenario	Description	Status
7	Follow Rent, no assets (FollowRent=true, Payments has only Amount)	Pending — FollowRent dropped from UI but API supports it
8	Follow Rent, 2 assets (FollowRent=true, assets without PaymentAmount)	Pending — same as above

13. Technical Notes & Gotchas

TeamDesk Token Behavior

- `<%= %>` auto-encodes: wraps strings in quotes, leaves numbers raw. Use for most column references.
- `<%=* %>` passthrough: inserts raw value. Required for:
 - Pre-computed JSON arrays (API Assembled RelatedAssets)
 - Boolean literals (API DoNotRelateAssets Value)
- `<=? %> ... <=?%>` conditional blocks: **DO NOT** reliably evaluate summary or formula columns. Avoid for API payloads.

Null Handling

- Use `Nz()` in formulas. If a field is null and concatenated with `&`, the entire expression returns null.
- `Concat()` treats nulls as empty strings — safer for JSON construction.
- Always validate that lookup columns resolve before submission.

Date Formatting

- Aspire expects `yyyy-MM-dd` (ISO format).
- TeamDesk `Format([Date], "yyyy-MM-dd")` produces the correct output.
- `XMLFormat([Date])` can also be used inside `JSONFormat()` for proper quoting.

Amount Formatting

- Send raw numeric values (no quotes, no currency symbols).
- `ToText([Fee Amount])` produces raw numeric in TeamDesk formula context.
- In Call URL body, `<%= %>` on a Numeric column outputs unquoted number.

Multi-Asset Grouping

- TeamDesk formulas cannot iterate child records.
- The solution uses a Summary column (`API Assembled RelatedAssets`) with Concatenate function to join all child Fee `[API Asset Fragment]` values with comma separator.
- The Call URL body wraps this in `[ ]` to form a valid JSON array.

14. Setup Links

Item	TeamDesk Path
Fees table columns	Setup > Tables > Fees > Columns
G&S table columns	Setup > Tables > Dealer Goods & Services > Columns
Call URL action	Setup > Tables > Dealer Goods & Services > Actions > Send Scheduled Billable to Aspire
Button configuration	Setup > Tables > Dealer Goods & Services > Buttons > Send Scheduled Billable Test
Equipment table	Setup > Tables > Miscellaneous Units > Columns